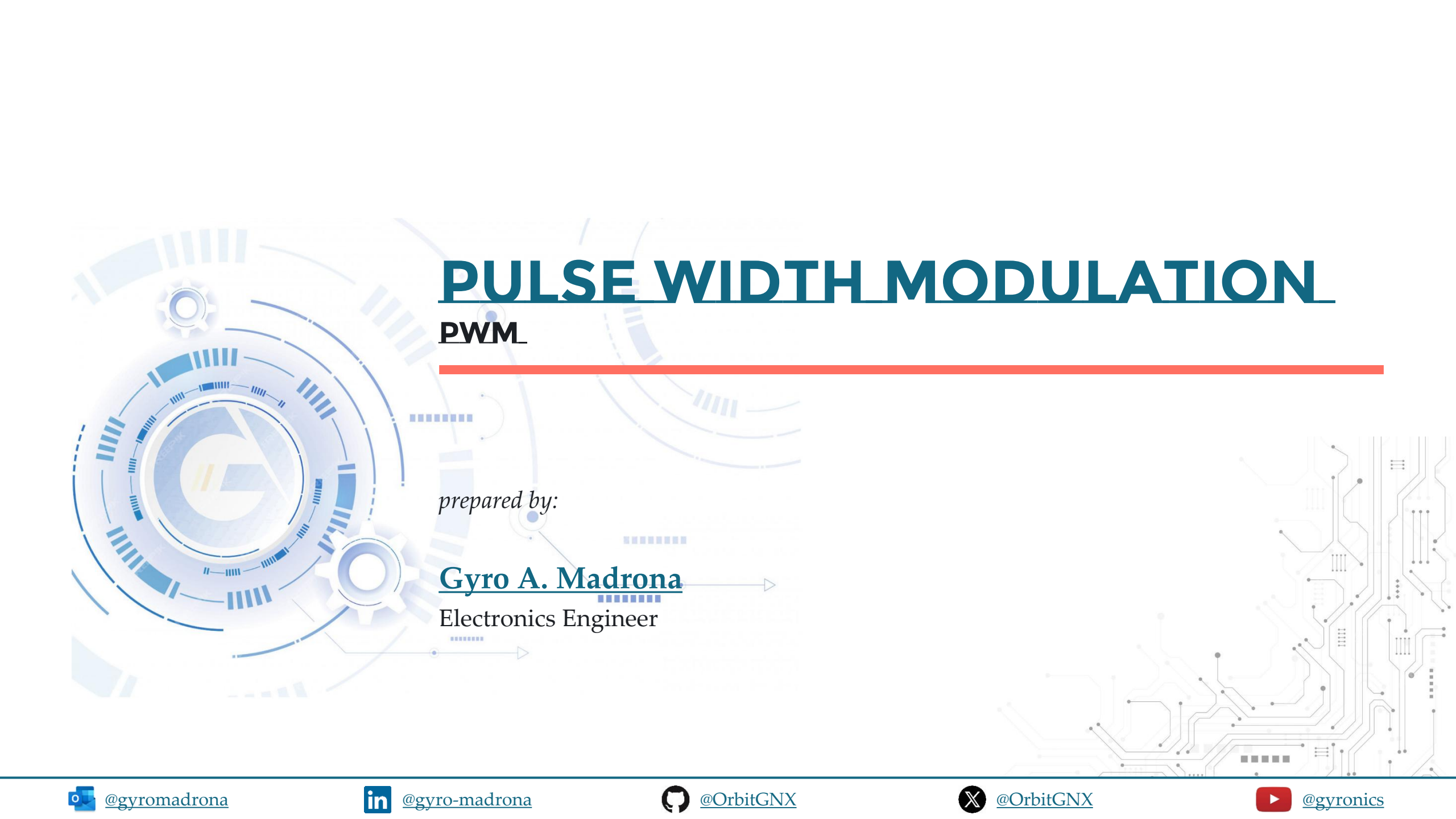




PULSE WIDTH MODULATION

PWM

prepared by:

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Electronics Engineer

TOPIC OUTLINE

Pulse Width Modulation (PWM)

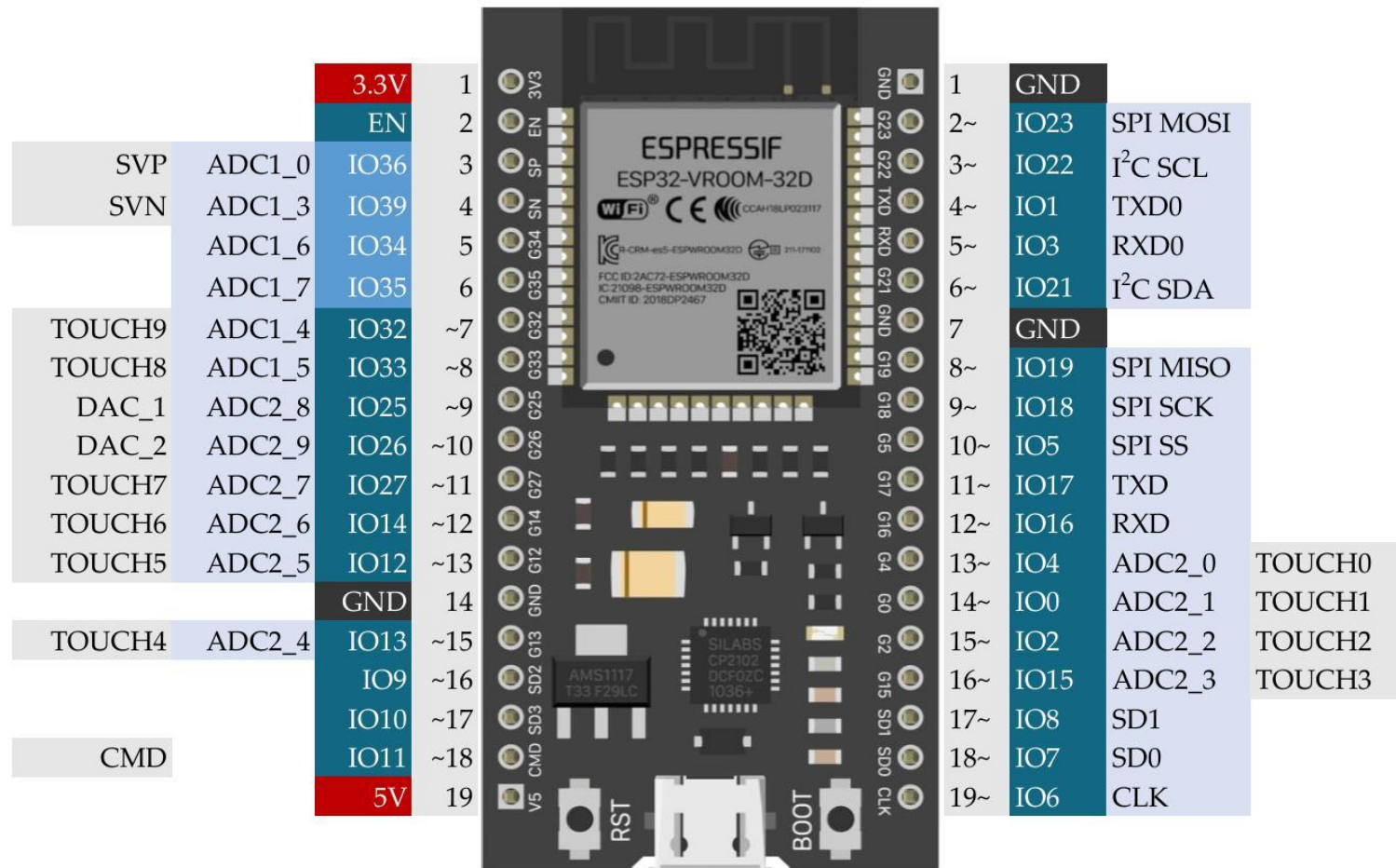
PWM Demo



PULSE WIDTH MODULATION



DEVKITC GPIOs MAP



GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

[ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation](#)

PULSE WIDTH MODULATION

Pulse Width Modulation (PWM) is a method of controlling power by changing the pulse width of a signal.

Parameters

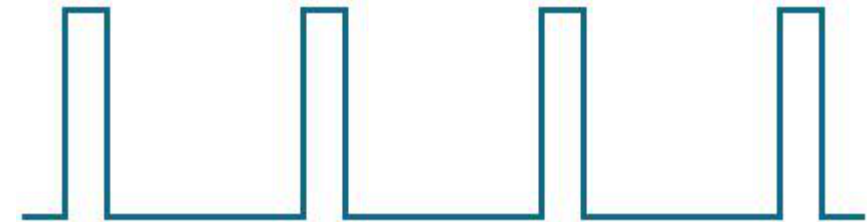
a. Duty cycle

Percentage of time the signal is “ON”

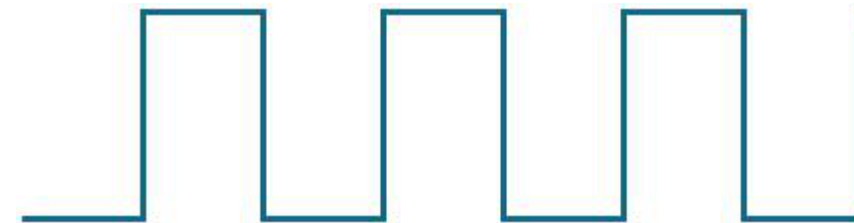
Average voltage = peak voltage · duty cycle

b. Switching frequency

20% duty cycle



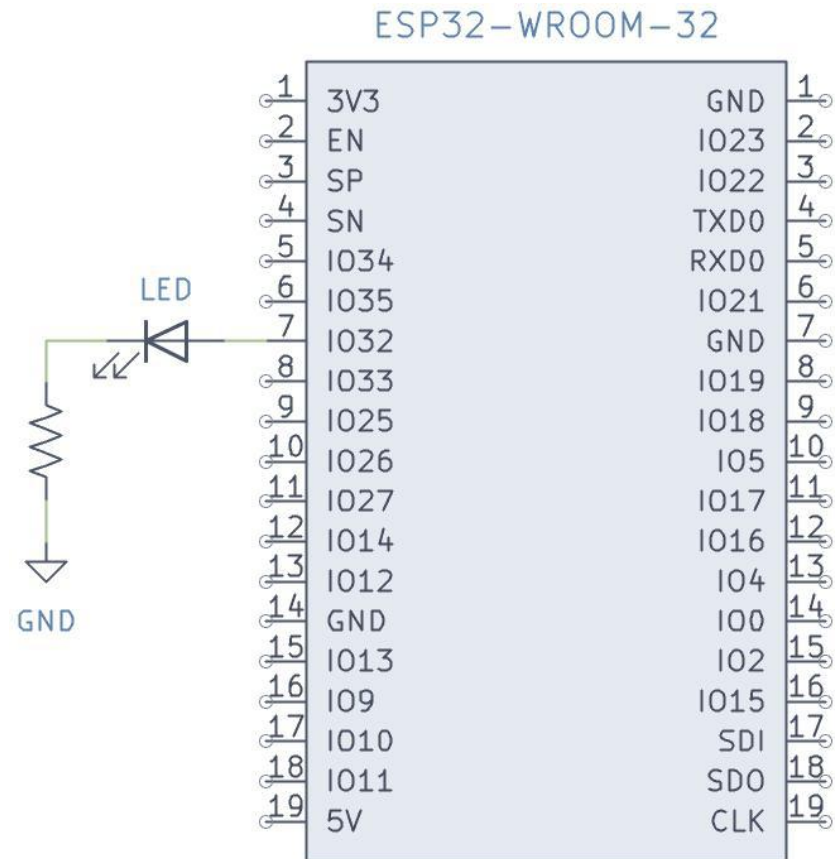
50% duty cycle



80% duty cycle



PWM DEMO



ledcAttach() – connect a GPIO pin to LEDC (LED Control) PWM hardware, allowing the pin to generate a PWM (Pulse Width Modulation) signal.

syntax

ledcAttach(pin, freq, resolution);

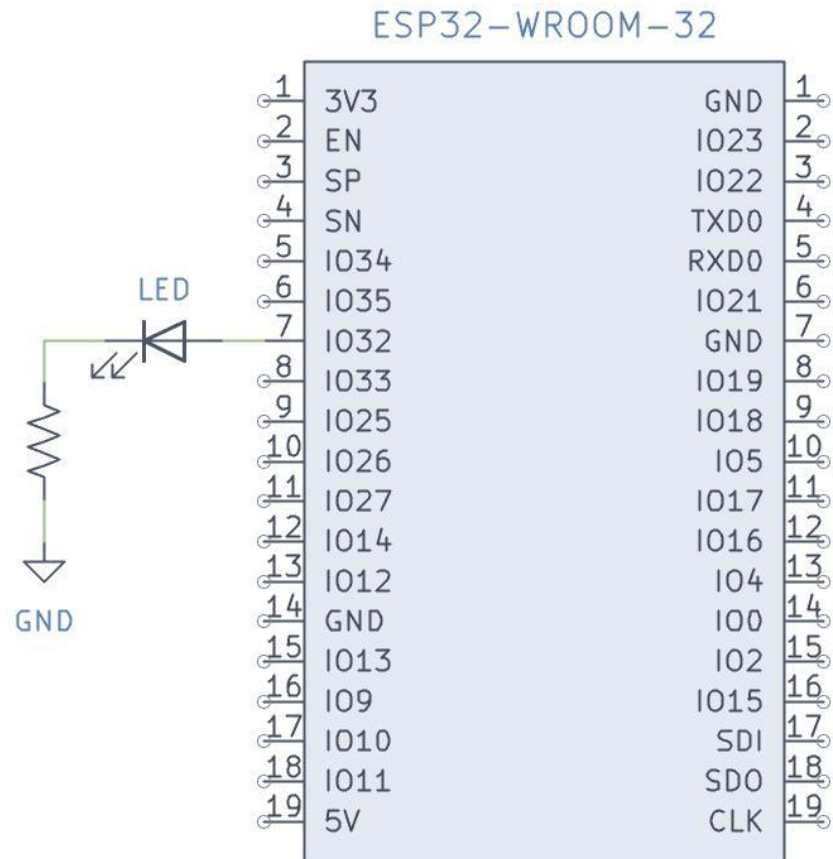
Resolution	Duty Cycle Range
8-bit	0 – 255
10-bit	0 – 1023
12-bit	0 – 4095

[LED Control \(LEDC\) - - — Arduino ESP32 latest documentation](#)

[arduino-esp32/cores/esp32/esp32-hal-ledc.h at master · espressif/arduino-esp32](#)



PWM DEMO



ledcWrite() – set the PWM duty cycle of a pin.

syntax

ledcWrite(pin, dutyCycle);

8-bit resolution

```
ledcWrite(5, 0);    // OFF
```

```
ledcWrite(5, 64);   // Dim, 25%
```

```
ledcWrite(5, 128);  // Medium, 50%
```

```
ledcWrite(5, 192);  // Bright, 75%
```

```
ledcWrite(5, 255);  // Max, 100%
```

[LED Control \(LEDC\) - - — Arduino ESP32 latest documentation](#)

[arduino-esp32/cores/esp32/esp32-hal-ledc.h at master · espressif/arduino-esp32](#)

LABORATORY

